

ESP32-Based Control System for ROV Navigation

System Overview

The first part of the system implements the **embedded control layer of the Remotely Operated Vehicle (ROV)** using an **ESP32 microcontroller**. This module is responsible for sensor acquisition, communication with the onboard computer, motor control, and safety management. The ESP32 acts as the real-time interface between the vehicle hardware and the higher-level control algorithms executed on the PC.

The control system integrates three sonar sensors for obstacle perception and two thrusters for horizontal propulsion. The ESP32 continuously reads the sonar measurements, transmits them to the PC for processing, and receives motion commands generated by the autonomous control algorithm.

Hardware Components

The embedded control system is composed of the following components:

- **ESP32 Microcontroller**
Serves as the main embedded controller responsible for sensor reading, serial communication, and motor control.
- **Sonar Sensors (3 units)**
Three sonar sensors are installed on the ROV to measure distances to nearby obstacles:
 - Front sonar sensor
 - Left sonar sensor
 - Right sonar sensor
- These sensors provide real-time distance measurements used to detect obstacles and assist navigation.
- **Thrusters (2 units)**
Two horizontal thrusters are used to control the movement of the ROV. These thrusters are controlled through Electronic Speed Controllers (ESCs) that receive PWM signals from the ESP32.
- **Electronic Speed Controllers (ESCs)**
ESCs convert PWM signals into motor power signals, allowing precise control of the thruster speed.
- **Onboard Computer (PC)**
The PC performs high-level decision making using machine learning models and sends motion commands to the ESP32.

System Architecture

The control architecture follows a **distributed control approach**:

- **ESP32**
 - Handles low-level hardware interaction
 - Reads sonar sensors
 - Controls thrusters
 - Enforces safety constraints
 - Communicates with the PC
- **PC**
 - Receives sensor measurements
 - Executes the control model
 - Computes navigation commands
 - Sends commands back to ESP32

This architecture separates **real-time hardware control** from **computationally intensive AI processing**, ensuring reliability and scalability.

Sensor Data Acquisition

The ESP32 continuously reads distance measurements from the three sonar sensors. These sensors provide the following measurements:

df : distance to the obstacle in front of the vehicle

dl : distance to the obstacle on the left side

dr : distance to the obstacle on the right side

To improve measurement stability, a **moving average filter** is applied to the sonar readings. This filtering process reduces noise and eliminates sudden spikes in sensor values.

After filtering, the ESP32 sends the measurements to the PC through the serial communication interface.

The transmitted data format is:

df,dl,dr

Example:

1.245,0.932,1.776

This format represents the filtered distance values measured in meters.

Serial Communication Protocol

The communication between the ESP32 and the PC is implemented using a **bidirectional serial communication protocol**.

Sensor Transmission (ESP32 → PC)

The ESP32 periodically sends sonar measurements to the PC. These measurements represent the current environmental perception of the ROV.

Format:

df, dl, dr

Where:

- df is the front sonar distance
- dl is the left sonar distance
- dr is the right sonar distance

Control Commands (PC → ESP32)

The PC sends motion commands to the ESP32 in the following format:

v, w

Example:

0.25,-0.40

Where:

- v represents the forward velocity command
- w represents the steering command

These commands are generated by the control algorithm running on the PC.

Thruster Control

The ESP32 converts the received motion commands into PWM signals for the thrusters.

The motion command consists of two components:

- **Forward velocity (v)**
Determines the forward thrust applied to both thrusters.
- **Steering command (w)**
Adjusts the relative thrust between the left and right thrusters to control turning.

The thruster commands are computed through a **differential thrust mechanism**:

- Left thruster command = forward velocity minus steering adjustment
- Right thruster command = forward velocity plus steering adjustment

This approach enables the ROV to perform forward movement and turning using only two horizontal thrusters.

The resulting commands are converted into PWM signals that control the ESCs connected to the thrusters.

Safety Mechanisms

A key feature of the ESP32 control module is the implementation of **fail-safe mechanisms** to ensure safe operation.

Command Timeout Protection

The ESP32 monitors the time since the last command received from the PC. If no command is received within a predefined timeout period (500 ms), the ESP32 immediately stops both thrusters.

This mechanism prevents uncontrolled movement in the event of communication failure or system errors.

Neutral Thruster State

When the system enters a fail-safe condition, both thrusters are set to their neutral PWM signal, which corresponds to zero thrust.

This ensures the ROV remains stationary until new commands are received.

Control Loop Operation

The ESP32 executes a continuous control loop at a fixed frequency of approximately **10 Hz**.

Each iteration of the control loop performs the following steps:

1. Read raw distance values from the sonar sensors.
2. Apply a moving average filter to reduce measurement noise.
3. Transmit the filtered sensor values to the PC.
4. Check for incoming motion commands from the PC.
5. Verify communication timeout conditions.
6. Convert the received commands into thruster PWM signals.
7. Update the thruster outputs accordingly.

This loop ensures real-time interaction between the ROV hardware and the higher-level control algorithms.

Role in the Autonomous System

The ESP32 module forms the **low-level control layer** of the ROV architecture. Its responsibilities include:

- Sensor data acquisition
- Hardware-level motor control
- Communication with the control computer
- Safety enforcement

Higher-level intelligence, including machine learning-based navigation strategies, is implemented on the PC. This layered architecture provides flexibility for integrating advanced control algorithms without compromising the stability of the embedded system.